

Judges Said

Demo video script — the full walkthrough

Target runtime **5:30** · two English prompts, one accepted and one refused · a 60-second cut and a full production kit are at the back.

What this video has to land in five minutes

- German court decisions are public domain, and reading them still costs four figures a year.
- Ask in English, search a German corpus, get real decisions — court, Aktenzeichen, date, paragraphs.
- It refuses things. Instantly, in code, before the model is ever called.
- It is built so that a hallucinated citation cannot reach the screen. That is the whole story.

Every number, quote and refusal message in this script is taken from the running system — the validator, the orchestrator, the glossary and the UI copy. Nothing here is invented for the camera. The two demo prompts are chosen so that their outcome is *deterministic*: the accepted one hits four glossary terms, the refused one matches a pattern that runs before retrieval.

00:00

Cold open

ON SCREEN

Black. An empty search box in the centre. One line of German types itself in, one character at a time, and stops. No cursor blink, no UI chrome yet.

TYPE

mein Arbeitgeber hat mir während der Probezeit während einer Krankschreibung gekündigt

VO

German, because that is the sentence the product was built for. Do not press Enter yet.

VO

“In Germany, every court decision is public domain. Every single one. Reading them still costs about a thousand euros a year.”

ON SCREEN

Hard cut to the logo on a plain background. One beat. Then cut to the app.

00:12

The problem, in twenty seconds

VO

“German case law lives behind Beck-online and juris. The people who need it most — someone who just got fired, an HR assistant at a five-person company, a works-council member — are exactly the people who cannot justify the licence.”

VO

“But § 5 of the Urheberrechtsgesetz removes copyright from statutes and court decisions completely. They are *amtliche Werke*. The evidence is free. Only the access is expensive.”

VO

“Judges Said closes that gap. Describe your situation in plain German or English, and get back the real decisions that match — with the court, the Aktenzeichen, the date, and the paragraphs the ruling actually turns on.”

B-ROLL

The README stat block, then the corpus table. Hold on the numbers for two seconds.

Corpus	Documents	Chunks
Court decisions (Arbeitsgerichtsbarkeit)	1,070	33,581
Statute sections	5,680	7,844
Total, embedded locally	6,750	41,425

00:35

The one rule

“Precedent, never predictions.”

VO “One rule shapes every design decision in this project: it shows precedent, never predictions.”

VO “That is not modesty. Under the Rechtsdienstleistungsgesetz, telling you what a court decided in a past case is *information*. Telling you how *your* case will turn out is a regulated legal service. So the app is built to refuse the second thing — and it refuses in code, not in a prompt. A prompt is not a compliance control.”

00:50

Demo 1 — the prompt that is accepted

ON SCREEN Signed in, empty thread, interface language set to English, dark or light — pick one and stay there. The sidebar shows a few earlier threads so the app does not look empty.

TYPE

My employer fired me during my probation period while I was on sick leave.

Four glossary hits: fired, probation period, sick leave, employer. Press Enter.

The line under the input is the shot

VO “The question is English. The corpus is German, and stays German. Watch the line under the input.”

APP SAYS

Searched in German for: Arbeitsunfähigkeit Krankschreibung Probezeit Arbeitgeber Kündigung

VO “Two searches just ran, not one. The semantic search got my original English — the embedding model is multilingual, so ‘fired while on sick leave’ already lands near ‘Kündigung während Arbeitsunfähigkeit’. No translation needed.”

VO “The keyword search cannot do that. A German full-text index will never match the token ‘dismissal’. So English gets turned into German search terms first — by a fixed labour-law glossary, not by a model. Free, instant, deterministic, and incapable of inventing a paragraph that does not exist. And it shows you what it searched for, because an English speaker with thin results deserves to see that ‘notice period’ became ‘Kündigungsfrist’.”

The status line

ON SCREEN Status chips replace each other: “Searching decisions...” → “Reading decisions...” → “Checking the answer...”

VO “Those labels are real stages, not a loading animation on a timer. There is a fourth one, ‘Revising the answer’, and you only ever see it if the answer actually failed a check and had to be redone. Nothing here is faked for effect.”

The answer, and the chips under it

ON SCREEN The answer appears as a whole, opening with one bold sentence, then one short paragraph per decision. Underneath: citation chips reading court · Aktenzeichen · date.

VO “Three decisions. Each chip is a real court, a real Aktenzeichen, a real date — and none of those three things were written by the language model. I will come back to that.”

ON SCREEN Click the first chip. The source panel slides in from the right.

VO “One click, and you are looking at the German passage the answer was built from. Not a summary of it — the passage itself, with its section: Tenor, Leitsatz, Tatbestand, Entscheidungsgründe. Underneath, the paragraphs the court applied. Click one of those and the statute text comes out of the same corpus.”

ON SCREEN Click a § chip — § 622 BGB is a good one. The statute text loads inside the panel.

The language toggle — the trust shot

ON SCREEN Switch the answer language from English to German on the same message. The prose changes. The chips underneath do not move.

VO “Switch the language and the prose is rewritten. The decisions do not change — because switching does not re-run the search. If it did, the German and the English version of the same answer could cite different cases, and that would destroy the only thing this product actually sells.”

VO

“And notice what never gets translated. ‘Arbeitsgericht Aachen’ does not become ‘Aachen Labour Court’. The Aktenzeichen, the date format, the paragraph symbols, the quoted passage — all German, in both languages. You might need to type that string into a court portal or read it to a lawyer. A translated quote is a paraphrase presented as evidence.”

02:05

Demo 2 — the prompt that is refused

ON SCREEN Same thread. Type slowly enough to read.

TYPE

Has anyone ever sued my employer for unfair dismissal?

Do not name a real company on camera. The trigger is the phrase, not the name.

ON SCREEN

The refusal appears immediately. No status chips. No spinner. No citation chips.

APP SAYS

I cannot answer that. Judges Said searches decisions by situation, not by person or company. Describe your situation instead and I will show comparable decisions.

VO

“Notice what did not happen. No spinner. No search. No model call. That question never reached the corpus at all.”

B-ROLL

Cut to **grounding/validator.py**, highlight **_PERSON_QUERY_PATTERNS**; then **chat/orchestrator.py**, highlight that the person-query branch is the *first* statement in **run_turn**, above retrieval.

VO

“That is a people-search product, not legal research. And German labour decisions carry health data, trade-union membership, sometimes religion — GDPR Article 9, the special categories. These decisions are pseudonymised, not anonymised. So person and employer queries are refused before retrieval, by a pattern check, deterministically, every time.”

VO

“And the refusal does the decent thing: it says what it cannot do, why, in one sentence, and offers what it can.”

Optional second refusal, if you want one more beat

Ask “**Will I win my case?**”. This one takes the long road: retrieval runs, the model answers, and the answer is checked. If it slips into prediction language it is rejected, the model gets exactly one more attempt, and if it fails again the user gets the refusal — *‘I cannot give an answer that is properly supported by the retrieved decisions. Predictions about how a case will turn out are out of scope.’* Powerful, but the exact wording depends on the model, so rehearse it before you record. The person-query refusal above is the one that is byte-for-byte identical every take.

02:45

The part that actually matters — hallucination

“A fake citation is not a bug here. It is the product failing at the one thing it exists to do.”

VO

“Now the part that actually matters.”

VO

“The known failure mode of a legal chatbot is not being wrong. It is being confidently wrong in a format that looks exactly right. A model that has read a million citations can produce a court, a file number and a date that are perfectly formatted and completely invented. Lawyers have been sanctioned in court for filing precisely that.”

VO

“So this project treats a hallucinated citation as the thing to design against. Four layers.”

Layer 1 — the citation metadata is never generated

VO

“Court name, Aktenzeichen, decision date, statutory references — none of those pass through the model as text it writes. They are stored database columns, looked up by the chunk ID the model cited. The model can choose a passage. It cannot spell a case number.”

Layer 2 — every quote must be verbatim in the German source

VO “Each quoted passage is checked against the original German text. Whitespace may differ, because a quote can be re-wrapped. One reworded word and it fails. Rejection code: `QUOTE_NOT_IN_SOURCE`. And the check runs against the German source no matter which language the answer is in — translating first would compare a paraphrase to a source, and pass anything.”

Layer 3 — you can only cite what you were given this turn

VO “If the answer cites a chunk that this turn’s retrieval never returned, it is rejected: `UNRETRIEVED_CITATION`. And an answer with no citations at all, that is not the honest ‘no comparable decision’ refusal, is rejected too: `NO_CITATIONS`. There is no path to an uncited claim.”

Layer 4 — the prediction guard

VO “Ten patterns, German and English: ‘Erfolgssaussichten’, ‘you will win’, ‘the court will rule in your favour’, ‘70 % Wahrscheinlichkeit’. Deliberately broad, and the trade-off is written into the code: a false positive costs one regenerated answer, a false negative is an unlicensed legal service.”

B-ROLL The rejection table from **docs/Architecture.md**, or the Rejection enum in `validator.py`. Five codes, one screen — this is the single most quotable frame in the video.

Rejection	Fires when
<code>PERSON_QUERY</code>	an aggregate person or employer search — checked <i>before</i> retrieval
<code>PREDICTION</code>	outcome-prediction language, German or English
<code>QUOTE_NOT_IN_SOURCE</code>	a quoted passage is not verbatim in the German source
<code>UNRETRIEVED_CITATION</code>	a cited chunk was not retrieved this turn
<code>NO_CITATIONS</code>	no citations, and not the explicit ‘no comparable case’ refusal

And then it fails closed

VO “Fail any of those and the model gets exactly one more attempt — told precisely which rule it broke. Fail again and the user gets a refusal, not an answer. It fails closed. An answer that never validated is never returned.”

VO “Which is why the answer is generated in full before a single character is streamed. Token-by-token looks nicer, but the check needs the complete text — and a prediction cannot be un-sent once somebody has read it. The endpoint still streams. It streams the *validated* text.”

B-ROLL Terminal: **uv run pytest**. Let the green line land on camera — 104 passed, about a second.

VO “And the whole validator is deterministic. No model, no network, no database. A hundred and four tests, about a second. Because a control that can only be verified by asking a model is not a control.”

04:00

Under the hood, fast

B-ROLL

The mermaid turn-lifecycle diagram from docs/Architecture.md, then quick cuts through retriever.py, fusion.py and query_language.py.

VO

“Retrieval is two legs. Each returns forty candidates. Their scores are not comparable — cosine similarity is bounded, the Postgres text rank is not — so they are fused by *rank*, not by score. Reciprocal Rank Fusion, k equals sixty.”

VO

“At most two passages survive per decision. Without that cap, one long on-topic judgment eats the entire answer and you see one case instead of the three to five the product promises.”

VO

“One measured finding I like: Postgres’ query builders AND their terms together, so every word of your question would have to appear inside one 350-token chunk. Measured — zero rows for nine questions out of ten. The keyword leg was silently contributing nothing and the fusion had quietly become semantic-only. Terms are OR-ed now, and the ranking function does the work it exists for.”

Piece	Choice	Why
Embeddings	multilingual-e5-base, 768-dim, run locally	cross-lingual by design; no embedding API, so the text never leaves the machine
Prefixes	query: on questions, passage: on indexed text	e5 degrades <i>silently</i> without them — the easiest mistake in the whole build
Lexical	tsvector with the german configuration	German stemming and compound splitting is why the keyword leg finds anything at all
Storage	one table, split by doc_kind	decisions and statutes rank against each other in a single query
Locator	named sections, no page column	Tenor, Leitsatz, Tatbestand, Entscheidungsgründe — they always exist, a page number does not
Agent	four bounded tools, no SQL	search again, re-read a chunk, read its neighbours, look up a statute — and nothing else

04:40

Shipping it

VO

“One number decided the entire deployment. The embedding model needs about 1.6 gigabytes of RAM. Every free platform tier caps at 512 megabytes.”

VO

“Shrinking it was tried properly — three different quantisation approaches: dynamic, per-channel, and static with calibration. All three fit in the budget. All three changed *which decisions got cited*, by twenty to a hundred percent. That is the one thing this product cannot get wrong, so quantisation was rejected on quality, not on effort.”

ON SCREEN

The deployment diagram from the README: browser → Render static site → Hetzner VPS (nginx + Let’s Encrypt) → Docker → Supabase.

VO

“So: a six-euro Hetzner box in Germany — which is also the right jurisdiction for a German legal-data product. Docker, nginx, Let’s Encrypt. The model is baked into the image, so a container start never waits on a one-gigabyte download, and it runs a single unicorn worker, because two workers would load two copies of the model. The frontend is a free static site. The database is Supabase Postgres with pgvector.”

VO

“Three things that break silently, and cost me an evening each:”

- “nginx buffers server-sent events by default, which makes streaming look exactly like a hang.”
- “The container has to bind to 127.0.0.1, or Docker’s iptables rules walk straight past the firewall and publish the API without TLS.”
- “And Supabase’s direct connection is IPv6-only. Without IPv6 on both the Docker daemon and the Compose network, the health check returns a cheerful 200 while every single query fails.”

05:15 **Close**

VO “Judges Said. It finds how German labour courts have actually ruled. It quotes them in German, verbatim, with one click to the passage. And when the corpus has nothing comparable, it says so — instead of inventing something that reads well.”

VO “Legal information. Not legal advice. Not a prediction.”

ON SCREEN End card: logo, the URL **judges-said-l49g.onrender.com**, and the one-line disclaimer — *Amtliches Werk, gemeinfrei nach § 5 UrhG. Keine Rechtsberatung.*

A The two prompts, ready to copy

Prompt 1 — accepted

TYPE

My employer fired me during my probation period while I was on sick leave.

Why it works, and why it is safe to record:

- Language detection reads it as English, so the answer defaults to English and the ‘Searched in German for’ line appears — that line is the whole bilingual story in one shot.
- It hits four glossary entries at once: **fired** → Kündigung, **probation period** → Probezeit, **sick leave** → Arbeitsunfähigkeit Krankschreibung, **employer** → Arbeitgeber. Both retrieval legs contribute, so the fusion is doing real work on camera.
- It is the same fact pattern as the German example in the README, so you can cut between the two languages and show the citations lining up.

Prompt 2 — refused

TYPE

Has anyone ever sued my employer for unfair dismissal?

Why this is the right refusal to demo:

- It is refused **before** retrieval and before any model call, so the refusal is instant and identical on every take — no model variance, nothing to re-shoot.
- It demonstrates the strongest compliance point in the project: the boundary is a deterministic check in the code path, not an instruction the model was politely asked to follow.
- It carries the GDPR Article 9 argument, which is the one that makes people sit up: labour decisions contain health data, union membership, sometimes religion.
- Use ‘my employer’, not a real company name — the trigger is the phrase, and naming a real employer in a video about refusing person-searches would be a bad look.

If you want a third prompt for the German half of the audience

“mein Arbeitgeber hat mir während der Probezeit während einer Krankschreibung gekündigt” — the cold-open sentence. Accepted, answers in German, and the ‘Searched in German for’ line stays hidden because no translation was needed. Good for showing that the German path is the shorter one.

B The 60-second cut

Same footage, ruthlessly trimmed. Use this for a post, a portfolio autoplay, or the top of the README.

Time	Shot	Voice-over
0:00–0:08	German sentence typing itself into the empty box	“Every German court decision is public domain. Reading them still costs a thousand euros a year.”
0:08–0:28	English prompt 1, the ‘Searched in German for’ line, three citation chips, one click into the German passage	“Ask in English. It searches a German corpus in German, and gives you the real decisions — court, Aktenzeichen, date, and the passage itself.”
0:28–0:40	English prompt 2, instant refusal, no spinner	“Ask it who sued whom and it refuses before it even searches. That check is code, not a prompt.”

Time	Shot	Voice-over
0:40–0:55	The rejection-code table, then the passing test run	"Quotes must be verbatim in the German source. Citations must come from what was actually retrieved. Outcome predictions are blocked outright. Fail the check twice and you get a refusal, not an answer."
0:55–1:00	End card with the URL	"Judges Said. Precedent, never predictions."

C Production kit

Before you hit record

- Sign in once and send both prompts. A cold container and a cold model make the first answer slow in a way that is not representative — the second run is the honest one.
- Seed three or four threads in the sidebar with real-looking titles, so the app does not look freshly installed.
- Browser at 125 % zoom, bookmarks bar hidden, no extension icons, notifications off. Record at 1080p or better and crop rather than shrink.
- Pick light or dark and stay there for the whole video. Toggling the theme on camera is a distraction unless the theme is the point.
- Never show `.env`, the Supabase dashboard, or a terminal with keys in the scrollback. If you show config, show `.env.example`.
- Have the answer scrolled fully into view before you click a citation chip — the panel opening over a half-scrolled message reads as a glitch.

B-roll shot list, in the order the script needs it

Cue	File	What to frame
Refusal	backend/app/grounding/validator.py	<code>_PERSON_QUERY_PATTERNS</code> , and the Rejection enum
Refusal	backend/app/chat/orchestrator.py	the person-query branch as the first statement of <code>run_turn</code>
Bilingual	backend/app/retrieval/query_language.py	the GLOSSARY dict — scroll it slowly, it reads well
Hallucination	docs/Architecture.md	the rejection table; hold it for three seconds
Hallucination	terminal	uv run pytest — the 104-passed line
Under the hood	docs/Architecture.md	the mermaid turn-lifecycle diagram, rendered
Shipping	README.md	the deployment stack diagram

Lower thirds / on-screen text

- **§ 5 UrhG** — court decisions carry no copyright
- **Searched in German for:** Arbeitsunfähigkeit · Probezeit · Kündigung
- **Refused before retrieval** — no search, no model call
- **GDPR Art. 9** — health, union membership, religion
- **Verbatim, or rejected**
- **Fails closed** — an unvalidated answer is never returned
- **104 tests. No model. No network. No database.**
- **Precedent, never predictions**

Two things to keep exactly as written

Say “**legal information, not legal advice**” out loud at least once, and never say the app tells you whether you will win — not even as a joke in the intro. The video is a public artefact about a product whose entire positioning is the RDG boundary; one throwaway line promising an outcome undoes the argument the rest of the video is making.

Pacing

The two demos are the only places where the pace should slow down. Everything else — the problem, the architecture, the deployment — can run fast over B-roll, because the viewer does not need to read it, only to register that it exists. If you have to cut for length, cut section 6 first and section 7 second. Never cut the hallucination section; it is the reason this video is worth making.